SERVICES OFFERED

PV Array Design & Optimization

Castillo Engineering delivers optimized PV array design — from stringing and setbacks to racking integration and equipment submittal review, we engineer layouts that maximize performance and streamline construction.

Development Engineering

Our goal at Castillo Engineering is to ensure you don't waste valuable time and money on unviable projects. We focus on upstream development, feasibility studies, grading vs. pile length analysis, preliminary drawings and interconnection support that ensure success. A one-stop-shop for engineering, we have honed our internal automation technologies to be more efficient - never losing sight of quality. With a deep understanding of regional and state needs, including utility interconnection requirements, we deliver designs that will meet project LCOE targets. We balance high quality designs and cost-effective service, managing risks to meet project milestones on time and within budget.

Electrical Engineering

Our electrical engineering services cover every critical aspect of your project, from high voltage, grid-tie design, data & monitoring systems interface, to module stringing, equipment BOMs, and thorough equipment submittal reviews-ensuring seamless integration and reliable performance from start to finish.

Civil Engineering

Full-scope civil engineering built for solar — from grading optimization and pile reveal mapping, to hydrologic and hydraulic (H&H) modeling, roadway, turnaround and laydown area designs, setbacks, and permit-ready IFC packages, we design with constructability, compliance, and efficiency in mind.

Substation Design

Comprehensive substation design from site prep and 3D layouts to system architecture, control panels, protection, equipment specs, power studies, and expert review—delivering a seamless solution from concept to completion.

Structural Engineering

We provide comprehensive structural engineering solutions, including structural calculation reports, equipment pads, racking foundation, tracker and above-ground cable management design. We deliver an expert equipment submittal review to ensure your project's success.

Power System Studies

Castillo includes in-depth power system studies that de-risk your project — from load flow and arc flash to grounding, harmonics, and relay settings — ensuring safe, stable, and compliant grid integration.

Battery Storage Design

End-to-end battery storage design — from system layout and interconnection to BOMs. We engineer energy storage that's ready to deploy.

Construction Support

Castillo enhances your construction projects with our comprehensive support services, including prompt responses to Requests for Information, thorough equipment submittal review and recommendations, and expert on-site evaluations and recommendations.

